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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

EmployeeAttendance Monitoring And Leave Management System

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Semester: **II**

DECLARATION

I hereby declare that theproject entitled “Employee Attendance Mointoring And Leave Management System” submitted in partial fulfillment of the requirements for the course Java programming is a **record of my original work** carried out under the guidance of **Aravind** Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.
I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____POOVIKA D_____

(Your Name)

CERTIFICATE

This is to certify that the projectentitled**"Employee Attendance Montoring And Leave Management System"** is a **bona fide work** carried out by Poovika D **I -'B'**, BSC Computer Science, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java programming**

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMBATORE

Date:10/04/2026

Submitted for the Viva-Voce Examination held on _____10/4/2026_____

Guide:
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ABSTRACT

The Employee Attendance Monitoring and LeaveManagement System is a Java-based application developed to manage and track employee attendance and leave records in an organized manner. The main objective of this system is to replace traditional manual attendance registers with an efficient automated solution.In many organizations, maintaining attendance and leave records manually leads to problems such as data errors, difficulty in tracking employee performance, time-consuming processes, and lack of proper record management.This project provides a computerized solution that allows employees to mark attendance, apply for leave, and enables administrators to monitor and manage records easily through a user-friendly interface.

The system is developed using Java and uses file handling or database concepts to securely store employee details. It provides features like employee login, attendance tracking, leave application, report generation, and secure data handling to ensure accuracy and reliability.

This project helps improve productivity, reduces manual work, saves time, and ensures quick access to employee records. The Employee Attendance Monitoring and Leave Management System can be used in offices, companies, institutions, and organizations for efficient workforce management.

Introduction

The Employee Attendance Monitoringand Leave Management System is a software application designed to manageandtrack employee attendance and leave records in an organized and efficient manner.Inmany organizations, maintaining attendance and leave details manually canleadtoissues such as data errors, misplacement of records, and difficulty in accessinginformation. This system provides a reliable and secure way to handle employeedatadigitally.

The system allows employeestomarktheir attendance, apply for leave, and view their attendance history easily.Italsoenables administrators to manage employee records, approve or reject leaverequests, and generate reports. By automating these processes, the system reducesmanual work, saves time, and improves overall productivity.

This project is developed using programming concepts such as Object-Oriented Programming (OOP) and data management techniques to handle employee information effectively. It demonstrates the practical use of software applications in real-world scenarios like offices, companies, and institutions.

Overall, the Employee Attendance Monitoring and Leave Management System helps in maintaining accurate records, improving data security, and simplifying workforce management processes.

1.1 Objective

•To develop a paperless attendance system
Replacemanual attendance registers withadigital system that records attendance electronically.

•To provide easy attendance tracking
Allowemployees to mark attendance andviewtheir attendance records quickly.

•To ensure data security and accuracy
Implementlogin and secure access to protectemployee information and avoid errors.

•To manage leave efficiently
Enableemployees to apply forleave andallowadmin to approve or reject requests easily.

•To improve productivity and reduce manual work
Savetime,reducepaperwork,andsimplifyattendanceand leave management processes.

1.2 Overview

TheEmployee Attendance Monitoring and Leave Management System is a Java-based software application designed to manage employee attendance and leave records in a digital format. It helps organizations maintain accurate records and replace traditional manual methods. The system ensures efficient tracking of employee presence and leave details.

In many organizations, managing attendance and leave manually is difficult, time-consuming, and prone to errors. This system provides a computerized solution that simplifies the process and improves efficiency. Employees can mark attendance, apply for leave, and view their records easily.

The system uses Java programming along with file handling or database concepts to store data securely. It includes features like user login, attendance tracking, leave management, and report generation. Overall, it reduces manual work, saves time, and improves productivity in organizations.

1.3 Features

User Authentication (Login & Registration)
Thesystem providessecureloginsothatonlyauthorized users (admin and employees) can access the system.

1. Attendance Marking Facility
Employees can mark their daily attendance, and the system records their presence accurately.

2. Leave Application
Employees can apply for leave easily by entering the required details through the system.

3. Leave Approval System
The admin can approve or reject leave requests based on availability and rules.

4. Attendance Tracking
The system keeps track of employee attendance including present, absent, and late records.

5. View Attendance and Leave Details
Employees can view their attendance history and leave status anytime.

6. Report Generation
The system generates reports for attendance and leave for better management and analysis.

7.Secure Data Storage
Allemployeerecordsarestored safely using file handling or database systems.

8.User-Friendly Interface
Thesystemisdesigned tobe simple and easy to use for both employees and administrators.

9.Time and Effort Saving
Reducesmanualwork, saves time, and improves overall efficiency of the organization.

1. PROGRAMANALYSIS

The Employee Attendance Monitoring and Leave Management System is developed using Java to provide a simple and efficient way to manage employee attendance and leave records digitally. The program follows a structured approach consisting of the user interface, application logic, and data storage. The user interface accepts inputs such as login details, attendance marking, and leave requests. The application logic processes these inputs and performs operations like validation, attendance recording, leave processing, and report generation. The data storage layer maintains employee details using Java file handling or a database such as MySQL or SQLite for permanent storage.

The system uses Object-Oriented Programming concepts like classes and objects to organize the program into modules such as the User Module, Attendance Module, and Leave Management Module. The user first logs into the system and selects the required operation from the menu. Based on the user's choice, the program processes the request and updates or retrieves the stored data. The system provides outputs in the form of attendance reports, leave status, and confirmation messages. Overall, the program reduces manual work, improves data accuracy, and ensures efficient employee management.

2.1 Problem Definition

Problem Definition – Employee Attendance Mointoring And Leave Management System

In many organizations such as offices, companies, schools, and institutions, employee attendance and leave records are still managed using traditional manual methods like registers or spreadsheets. This manual system creates several problems including errors in data entry, difficulty in maintaining large records, and time-consuming processes. Tracking employee attendance and leave status becomes complicated, and generating reports requires extra effort. There is also a risk of data loss, duplication, and lack of proper security.

Finding specific attendance or leave information takes time, and managing multiple employees becomes difficult. The manual system also increases administrative workload and

reduces overall efficiency. There is a strong need for a computerized solution that can manage attendance and leave records accurately and quickly.

2.2 Project Overview

Project Overview – Employee Attendance Montoring And Leave Management System (Java)

The Employee Attendance Monitoring and Leave Management System is a Java-based application developed to manage employee attendance and leave records in a digital and organized manner. The main purpose of this project is to replace traditional manual record keeping with a computerized system that allows easy tracking, storage, and management of employee data.

This project provides features such as user login authentication, attendance marking, leave application, leave approval, report generation, and record management. The system ensures that only authorized users can access the data, thereby maintaining security and privacy. It uses Java programming concepts such as object-oriented programming, file handling, and optionally database connectivity to store and manage employee details effectively.

2.3 Scopes and Limitations

Scope

The Employee Attendance Monitoring and Leave Management System has a wide scope in organizations where employee management is essential. The system can be used in offices, companies, schools, colleges, and institutions to manage attendance and leave records digitally. It allows employees to mark attendance, apply for leave, and view records easily, while administrators can manage employee data and generate reports. This reduces manual work and improves efficiency. The project demonstrates the use of Java programming, file handling, and database connectivity to build a secure and user-friendly system. The system can be further enhanced by adding features like mobile access, biometric integration, cloud storage, and real-time notifications in the future.

Limitations

Although the system provides basic attendance and leave management functions, it has some limitations. The project is mainly designed for small-scale use and may not handle very large organizations with huge data effectively. Advanced features such as biometric attendance, cloud backup, and high-level security are not included in the current version. The system may require manual data backup to prevent data loss, and performance may depend on the system where it is installed. These limitations can be improved in future versions by integrating advanced technologies.

2.SYSTEM DESIGN

System Design – Digital Document Record Management System (Java)

The Employee Attendance Monitoring and Leave Management System is designed using a three-layer architecture consisting of the presentation layer, application layer, and data storage layer. The presentation layer provides a user-friendly interface where users can log in and perform operations such as marking attendance, applying for leave, and viewing records. The application layer contains the business logic that processes user requests, validates inputs, and manages attendance and leave operations. The data storage layer is responsible for storing employee details, attendance records, and leave information using Java file handling or a database such as MySQL or SQLite.

The system is divided into modules including the User Authentication Module, Attendance Module, Leave Management Module, and Report Module. The leave management module manages leave applications, approvals, and leave balance. The report module generates attendance and leave reports for analysis. The system collects inputs such as user details and attendance data, processes the information, and provides outputs like attendance reports, leave status, and confirmation messages. This design makes the system simple, secure, and easy to maintain and expand.

3.1 Module

1. UserAuthenticationModule

This module manages user login and verification. It checks the username and password before allowing access to the system. Only authorized users (admin and employees) can use the system, ensuring data security and privacy.

2. Attendance Module

This module allows employees to mark their daily attendance. It records details such as login time, logout time, and attendance status. It helps in maintaining accurate attendance records.

3. Leave Management Module

This module enables employees to apply for leave. It allows the admin to approve or reject leave requests. It also keeps track of leave balance and leave history.

4. Employee Management Module

This module manages employee details such as adding, updating, and viewing employee information. It helps in maintaining organized employee records.

5. Storage Module

This module handles storing and retrieving employee data. It uses Java file handling or a database to store attendance and leave details securely.

6. Report/View Module

Thismoduledisplays attendance reports, leave status, and employee details. It shows the results of operations like attendance marking, leave approval, and data updates.

3. METHODOLOGY

The Employee Attendance Monitoring and Leave Management System is developed using a structured and step-by-step methodology to ensure the system is simple, secure, and efficient. The development process begins with requirement analysis, where the problems of manual attendance and leave management are studied. Based on these requirements, the system design is prepared by identifying modules such as user authentication, attendance tracking, leave management, and reporting.

After the design phase, the implementation phase is carried out using the Java programming language. Object-Oriented Programming concepts such as classes, objects, methods, and encapsulation are used to develop different modules. Java file handling or database connectivity is used to store employee details, attendance records, and leave information securely.

Once the system is implemented, testing is performed to ensure all functions like login, attendance marking, leave application, approval, and report generation are working correctly. Errors are identified and corrected during this stage. Finally, the system is deployed for use, allowing organizations to manage employee attendance and leave digitally in an efficient and systematic manner.

4.1. Problem Analysis

Problem Analysis – Employee Attendance Monitoring And Leave Management System

Traditional attendance and leave management in many organizations still depends on manual methods such as registers or paper records. This approach creates several challenges in recording, organizing, and maintaining employee data efficiently. Managing large volumes of attendance records requires more time and effort, and it becomes difficult as the number of employees increases. There is also a risk of errors, data loss, or duplication due to manual handling.

Tracking employee attendance and leave manually is time-consuming and reduces productivity. Finding specific attendance records or leave details takes more time and effort. Sharing information between departments or management is also slow and inconvenient. In

addition, manual systems lack proper security, as unauthorized access or changes to records may occur.

Maintaining and updating attendance and leave records increases administrative workload and operational costs. Therefore, there is a need for an automated system that can manage employee attendance and leave efficiently, improve accuracy, and ensure secure and easy access to data.

4.2. System Design

System Design – Employee Attendance Monitoring And Leave Management System (Java)

The Employee Attendance Monitoring and Leave Management System is designed as a structured and modular application to ensure easy usage, security, and scalability. The system follows a three-layer architecture consisting of the presentation layer, application layer, and data storage layer. The presentation layer provides a simple menu-driven interface where users can log in and perform operations such as marking attendance, applying for leave, and viewing records. The application layer acts as the core of the system, processing user requests, validating inputs, and controlling all attendance and leave-related operations. The data storage layer stores employee details, attendance records, and leave information using Java file handling or a database such as MySQL or SQLite. The system is divided into major

modules including user authentication, attendance
management, leave management, employee management, and storage. The user authentication module verifies login credentials to ensure only authorized users can access the system. The attendance module allows employees to mark daily attendance. The leave management module enables employees to apply for leave and allows admin to approve or

reject requests. The employee management module handles employee data, and the storage module organizes and saves all records securely.

The system accepts inputs such as user credentials, attendance data, and leave details, processes them using Java logic, and provides outputs like attendance reports, leave status,

and confirmation messages. This design ensures the system is efficient, secure, and easy to maintain.

4.3. Program Development

The Employee Attendance Monitoring and Leave Management System was developed using Java by following a structured development process. Initially, system requirements were analyzed to identify the problems in manual attendance and leave management. Based on these requirements, the system was designed by dividing it into modules such as user authentication, attendance tracking, leave management, and employee management. Each module was planned carefully to ensure smooth system operation.

After designing the system, coding was implemented using Object-Oriented Programming concepts in Java. Classes, objects, methods, and encapsulation were used to develop the modules effectively. F Errors were identified and corrected during testing. Finally, the system was completed and made ready for use as an efficient and reliable solution.

4.4. Testing and Validation

The system was tested to ensure all functions such as login, attendance marking, leave application, approval, and report generation work correctly. Different test cases were used to check the performance of each module. Errors identified during testing were corrected to improve system reliability. Validation confirmed that the system meets user requirements and provides accurate results. It also ensures that employee data is stored securely and processed correctly.

4.5. Result and Documentation

Result:

The Employee Attendance Monitoring and Leave Management System was successfully developed using Java. The system allows employees to mark attendance, apply for leave, and view their records easily. It helps administrators manage employee data, approve or reject

leave requests, and generate reports. The system reduces manual work, improves accuracy, and increases efficiency in managing attendance and leave. It provides a user-friendly interface and ensures secure access through authentication.

Documentation:

Theproject documentation includes all stages of development such as problem definition, objectives, system design, methodology, program development, testing, and results. It explains how the system works, the technologies used, and the features provided. The documentation serves as a guide for users and future developers to understand, maintain, and enhance the system.

4.6. Algorithm

Start the system

- Initialize variables and load employee data
- Display login page
- Enter username and password
- Validate user credentials
- If valid, display main menu
- Select option (Mark Attendance / Apply Leave / View Records)
- Process the selected operation
- Store or update data in file/database
- Display result (attendance status or leave details)

- Repeat steps if required
- Stop the system

4.7. Flowchart

```
+----------------------+
|       START        |
+----------------------+
        |
        v
+----------------------+
|  Initialize  System  |
+----------------------+
        |
        v
+----------------------+
|    Login Page        |
+----------------------+
        |
        v
+----------------------+
| Enter Username & Pwd |
+----------------------+
        |
        v
+----------------------+
| Validate Credentials |
+----------------------+
```

```
        |
    +----+----+
    |         |
v         v

+---------+ +----------------------+
| Invalid | |       Valid          |
+---------+ +----------------------+
    |              |
    v              v
+----------------+ +----------------------+
| Error Message | | Display Menu          |
+----------------+ +----------------------+
    |              |
    +------<--------+
                   |
                   v
      +---------------------------+
      | 1.Attendance 2.Leave      |
      | 3.View 4.Exit             |
      +---------------------------+
                   |
                   v
      +---------------------------+
      | Select Option             |
      +---------------------------+
                   |
    +---------------+---------------+
    |              |               |
```

```
    v             v             v
+-------------++----------------+ +------------------+
|Attendance||ApplyLeave         | | View Records     |
+-------------++----------------+ +------------------+



|             |             |
    v             v             v
+-------------++----------------+ +------------------+
|MarkPresent||EnterDays          | | Fetch Data       |
+-------------++----------------+ +------------------+
    |             |             |
    v             v             v
+-------------++----------------+ +------------------+
|SaveData     ||CheckBalance | | Display Records |
+-------------++----------------+ +------------------+
                  |
            +------+------+
            |            |
            v            v
       +-------------+ +----------------+
       |Approved    ||NotApproved |
       +-------------+ +----------------+
            |            |
            v            v
       +-------------+ +----------------+
       |UpdateData||Show Message |
       +-------------+ +----------------+
                  |
```

```
            v
        +----------------------+
        | Back to Menu         |
        +----------------------+

  |
            v
        +----------------------+
        |       EXIT           |
        +----------------------+
            |
            v
        +----------------------+
        |        END           |
        +----------------------+
```

4. **SOURCE CODE**

```
import java.util.*;

class Employee {
    int id;
 String name;
    int totalDays = 0;
    int presentDays = 0;
    int leaveDays = 0;

 Employee(int id, String name) {
```

```
        this.id = id;
        this.name = name;
    }



    voidmarkAttendance(boolean isPresent) {
        totalDays++;
        if(isPresent) {
            presentDays++;
        }else {
            leaveDays++;
        }
    }

    voidapplyLeave(int days) {
        leaveDays += days;
        totalDays += days;
    }

    voiddisplay() {
        System.out.println("\nEmployee ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Total Days: " + totalDays);
        System.out.println("Present Days: " + presentDays);
        System.out.println("Leave Days: " + leaveDays);
    }
}

public class AttendanceSystem {
```

```
staticScanner sc = new Scanner(System.in);
staticArrayList<Employee> empList = new ArrayList<>();



// Add Employee
staticvoidaddEmployee() {
    System.out.print("Enter Employee ID: ");
    int id = sc.nextInt();
    sc.nextLine();

    System.out.print("Enter Employee Name: ");
    Stringname = sc.nextLine();

    empList.add(new Employee(id, name));
    System.out.println("Employee Added Successfully!");
}

// Find Employee
staticEmployee findEmployee(int id) {
    for(Employee e : empList) {
        if(e.id== id) {
            return e;
        }
    }
    return null;
}

// MarkAttendance
```

```
staticvoidmarkAttendance() {
    System.out.print("Enter Employee ID: ");
    int id = sc.nextInt();



    Employee e = findEmployee(id);
    if(e!=null) {
        System.out.print("Is Present (true/false): ");
        boolean present = sc.nextBoolean();
        e.markAttendance(present);
        System.out.println("Attendance Updated!");
    } else {
        System.out.println("Employee Not Found!");
    }
}

// Apply Leave
staticvoidapplyLeave() {
    System.out.print("Enter Employee ID: ");
    int id = sc.nextInt();

    Employee e = findEmployee(id);
    if(e!=null) {
        System.out.print("Enter number of leave days: ");
        intdays = sc.nextInt();
        e.applyLeave(days);
        System.out.println("Leave Applied!");
    } else {
        System.out.println("Employee Not Found!");
```

```
        }
    }

// Display Employee Details

    static void displayEmployee() {
        System.out.print("Enter Employee ID: ");
        int id = sc.nextInt();

        Employee e = findEmployee(id);
        if (e != null) {
            e.display();
        } else {
            System.out.println("Employee Not Found!");
        }
    }

// Display All Employees
    static void displayAll() {
        for (Employee e : empList) {
            e.display();
        }
    }

public static void main(String[] args) {

        int choice;

        do {
```

```
        System.out.println("\n===== Employee Attendance System =====");
        System.out.println("1. Add Employee");
        System.out.println("2. Mark Attendance");
        System.out.println("3. Apply Leave”);
System.out.println("4. Display Employee");
        System.out.println("5. Display All Employees");
        System.out.println("6. Exit");

        System.out.print("Enter your choice: ");
        choice = sc.nextInt();

        switch (choice) {
            case 1:
                addEmployee();
                break;
            case 2:
                markAttendance();
                break;
            case 3:
                applyLeave();
                break;
            case 4:
                displayEmployee();
                break;
            case 5:
                displayAll();
                break;
            case 6:
                System.out.println("Thank You!");
```

```
                break;
            default:
System.out.println("Invalid Choice!");

        }

    } while (choice != 6);
  }
}
```

4. INPUT AND OUTPUT SCREENS

INPUT SCREEN

```
1.Add Document
2.View Documents
3.Exit
```

OUTPUT SCREEN

===== Employee Attendance System =====

1. Add Employee

2. Mark Attendance

3. Apply Leave

4. Display Employee

5. Display All Employees

6. Exit

Enter your choice: 1

Enter Employee ID: 101

Enter Employee Name: Ravi

Employee Added Successfully!

===== Employee Attendance System =====

1. Add Employee

2. Mark Attendance

3. Apply Leave

4. Display Employee

5. Display All Employees

6. Exit

Enter your choice: 1

Enter Employee ID: 102

Enter Employee Name: Priya

Employee Added Successfully!

===== Employee Attendance System =====

Enter your choice: 2

Enter Employee ID: 101

Is Present (true/false): true

Attendance Updated!

===== Employee Attendance System =====

Enter your choice: 2

Enter Employee ID: 102

Is Present (true/false): false

Attendance Updated!

===== Employee Attendance System =====

Enter your choice: 3

Enter Employee ID: 102

Enter number of leave days: 2

Leave Applied!

===== Employee Attendance System =====

Enter your choice: 4

Enter Employee ID: 101

Employee ID: 101

Name: Ravi

Total Days: 1

Present Days: 1

Leave Days: 0

===== Employee Attendance System =====

Enter your choice: 4

Enter Employee ID: 102

Employee ID: 102

Name: Priya

Total Days: 3

Present Days: 0

Leave Days: 3

===== Employee Attendance System =====

Enter your choice: 5

Employee ID: 101

Name: Ravi

Total Days: 1

Present Days: 1

Leave Days: 0

Employee ID: 102

Name: Priya

Total Days: 3

Present Days: 0

Leave Days: 3

===== Employee Attendance System =====

Enter your choice: 6

Thank You!

5.DISCUSSION & CONCLUSION

7.1 Discussion

TheEmployee Attendance Monitoring and Leave Management System was developed to replace manual attendance and leave handling with a simple digital solution. The system successfully performs basic operations such as marking attendance, applying for leave, viewing records, and managing employee data. It improves record organization, reduces paperwork, and provides faster access to attendance and leave details.The project also demonstrates the practical use of Java programming concepts such as file handling, arrays or collections, and menu-driven programming.

7.2 Conclusion

The Employee Attendance Monitoring and Leave Management System successfully replaces manual methods with an efficient digital solution. It helps in managing employee attendance and leave records accurately and quickly. The system reduces paperwork, minimizes errors, and saves time for both employees and administrators.By using Java programming concepts, the project demonstrates how real-world problems can be solved using software applications.

6. REFERENCES

References

1.Herbert Schildt – *Java: The Complete Reference* (used for core Java concepts and programming techniques)

2.E. Balagurusamy – *Programming in Java* (helped in understanding OOP concepts and program structure)

3.Oracle official Java documentation (used for syntax, libraries, and standard methods)

4.GeeksforGeeks (used for examples, logic building, and problem-solving approaches)

5.W3Schools (used for basic programming concepts and simple explanations)

6.Various online tutorials and study materials (used for project guidance, coding support, and implementation ideas)